Warm-up | Before Section 3.8

Two AP-style multiple-choice questions | Show work and select one answer for each.

1 **NO CALCULATOR**

Recent Review | 3.7 Implicit Differentiation

For the curve $x^2 + xy + y^2 = 7$ what is the slope of the tangent line at (2, 1)?

A. $-\frac{5}{4}$

B. $-\frac{4}{5}$

C. $\frac{5}{4}$

D. $\frac{4}{5}$

E. $-\frac{3}{4}$

WORKSPACE

2 **CALCULATOR PERMITTED**

Spiral Review | 3.4 Velocity and Rates

A particle has position $s(t) = t^2 \sin t$ meters. What is its instantaneous velocity at t = 1.4 s?

Round to the nearest hundredth; angles are in radians.

A. 1.07 m/s

B. 2.76 m/s

C. 3.09 m/s

D. 3.57 m/s

E. 4.24 m/s

WORKSPACE

Derivatives of Inverse Trigonometric Functions

01 **Find the derivative of**

$$h(x) = x\sec^{-1}(3x)$$

Derivatives of Inverse Trigonometric Functions

02 **Find the derivative of**

$$f(x) = x\sqrt{1-x^2} + \cos^{-1}(x^3)$$

Derivatives of Inverse Trigonometric Functions

03 **Find an equation for the tangent line to the graph of**

$$y = \tan^{-1} x \quad \text{at} \quad (1, \pi/4)$$

Derivatives of Inverse Trigonometric Functions

04 **Find the inverse-function derivative**

$$(f^{-1})'(2) \quad \text{if} \quad f(x) = x^3 + 2x - 1$$

Derivatives of Inverse Trigonometric Functions

05 **Find the inverse-function derivative**

$$(f^{-1})'(6) \quad \text{if} \quad f(x) = x^3 - \frac{2}{3}$$

Derivatives of Inverse Trigonometric Functions

06 **Find the derivative of y with respect to the appropriate variable.**

$$y = \cos^{-1}(x^2)$$

Derivatives of Inverse Trigonometric Functions

07 **Find the derivative of y with respect to the appropriate variable.**

$$y = \sin^{-1}\left(\frac{3}{t^2}\right)$$

Derivatives of Inverse Trigonometric Functions

08 **A particle moves along the x-axis, with $t \geq 0$. Find its velocity at $t = 1$.**

$$x(t) = \tan^{-1}(t^2), \quad t = 1$$

Derivatives of Inverse Trigonometric Functions

09 **Find the derivative of y with respect to the appropriate variable.**

$$y = \sec^{-1}(2s + 1)$$

Derivatives of Inverse Trigonometric Functions

10 **Find the derivative of y with respect to the appropriate variable.**

$$y = \csc^{-1}(x^2 + 1), \quad x > 0$$

Derivatives of Inverse Trigonometric Functions

11 **Find the derivative of y with respect to the appropriate variable.**

$$y = \sqrt{s^2 - 1} - \sec^{-1}(s)$$

Derivatives of Inverse Trigonometric Functions

12 **Find an equation for the tangent line to the graph at x = 2.**

$$y = \sec^{-1} x, \quad x = 2$$

Derivatives of Inverse Trigonometric Functions

13 **Find an equation for the tangent line to the graph at x = 3.**

$$y = \sin^{-1}\left(\frac{x}{4}\right), \quad x = 3$$